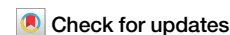


<https://doi.org/10.1038/s41534-025-01106-w>

Reducing quantum resources for attacking S-AES on quantum devices



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The Advanced Encryption Standard (AES) is one of the most widely used symmetric block cipher cryptography algorithms. Simplified-AES (S-AES) retains the structural framework of AES, making it an ideal model for elucidating the mechanism of symmetric cryptography. In this paper, we investigate potential quantum attacks on S-AES, utilizing both Grover-based algorithms and variational quantum algorithms (VQAs). For Grover-based attacks, we introduce an optimized oracle for implementing S-AES, reducing the complexity from 160 Toffoli gates to 120 by refining the SubNibble operation. Additionally, we propose a general Variational Quantum Attack Algorithm (VQAA) for any type of symmetric encryption that does not require the quantum implementation of the encryption procedure, which reduces the gate complexity substantially. These improvements enable the implementation of both Grover-based attacks and VQAA on superconducting quantum computers. The results demonstrate the feasibility of cryptographic analysis on currently available quantum hardware. Our findings also underscore the challenges of parameter optimization for S-AES, attributing the difficulty to the design of efficient parameterized quantum circuits, which constrain the effectiveness of variational algorithms. Despite these advancements, significant challenges remain in achieving quantum attacks on AES.

One of the most pivotal applications of quantum algorithms is cryptanalysis. Asymmetric cryptography, exemplified by RSA¹, relies on hidden subgroup problems, which Shor's algorithm can efficiently break in polynomial time^{2–4}. Additionally, Grover's algorithm^{5–7} offers quadratic speedup in search problems compared to classical algorithms, making it instrumental for attacking symmetric cryptography. Consequently, numerous studies^{8–12} have focused on Grover's algorithm against AES, the foremost symmetric cryptography standard. The primary objective of these studies is to construct the oracle in Grover's attack, which maps plaintext and key to ciphertext according to AES encryption. For example, AES-128 uses a 128-bit key to encrypt 128 bits of plaintext, reaching a practical scale in cryptography. These works^{8–12} reveal that each oracle comprises approximately $\mathcal{O}(10^5)$ logical gates, necessitating $\mathcal{O}(2^{64})$ iterations. Consequently, the implementation of these algorithms remains distant from near-term quantum devices.

S-AES¹³ shares the same structural framework as AES but operates on 16 bits instead of 128 bits. The encryption process encompasses four operations: AddRoundKey, SubNibbles, ShiftRows, and MixColumns. To attack S-AES, Grover's algorithm requires approximately $\mathcal{O}(2^8)$ iterations,

and each oracle uses fewer gates compared to full AES. This makes executing the Grover-based attack for S-AES and demonstrating its quantum speedup in the near future is feasible. In 2018, Almazrooie et al.¹⁴ proposed a quantum implementation of S-AES, utilizing 72 qubits, 192 Toffoli gates, and hundreds of Clifford gates. Notably, in 2021, several enhanced quantum implementations of S-AES^{10,15} reduced the quantum resources to 32 qubits, $16 \wedge_3 X$ gates (or CCCX gate), 96 Toffoli gates, and hundreds of Clifford gates. In addition, the parameterized quantum circuit^{16–21} is an alternative and competitive approach for implementing quantum attacks on near-term quantum devices, owing to its shallow circuit depth. In 2022, Wang et al.²² introduced a variational quantum attack algorithm (VQAA) tailored for symmetric cryptography and validated its efficacy on the Simplified-Data Encryption Standard (S-DES).

In this paper, we first present an efficient quantum implementation of the S-AES within a Grover-based attack framework, significantly reducing the overall quantum resource costs. Subsequently, we simulate the Grover-based attack on S-AES using Qiskit²³, achieving the expected results. Furthermore, we introduce a Variational Quantum Attack Algorithm (VQAA) that circumvents the need for the quantum implementation of symmetric

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encryption procedures on quantum computers, while preserving computational efficiency. Utilizing these approaches to optimize quantum resources, we successfully implement a modular quantum version of S-AES and the VQAA on superconducting quantum computers available via the IBM platform.

Results

Quantum S-AES Oracle

Grover-based algorithms attack S-AES by searching the target item in an unstructured database containing N items. Initially, the quantum state is prepared in a superposition state $|\psi_0\rangle = (1/\sqrt{N}) \sum_{x=0}^{N-1} |x\rangle$. After $\frac{\pi}{4} \sqrt{\frac{N}{M}}$ times Grover iterations, where M is the number of target items in the database, the probability of observing the target state(s) reaches $P = 1 - \epsilon$, where ϵ is the false probability and decreases exponentially with the number of qubits. The Grover iteration operator is defined as $G = (2|\psi_0\rangle\langle\psi_0| - I)O$, which amplifies the amplitude of the target state $|x_0\rangle$. Here, oracle O maps $|x\rangle$ to $(-1)^{f(x)}|x\rangle$, where the function $f(x) = 0$ if $x \neq x_0$ and $f(x) = 1$ if $x = x_0$. Thus, for Grover-based algorithms, the key point is to construct the quantum S-AES Oracle that maps $|x\rangle$ to $(-1)^{f(x)}|x\rangle$. Specifically, in the context of attacking S-AES, the oracle O flips the phase of the state $|K_0\rangle$ while leaving other quantum states unchanged, where $|K_0\rangle$ corresponds to the correct key K_0 in S-AES. Given a plaintext P and a key K , the S-AES encryption algorithm produces the ciphertext C through the mapping $C = g_P(K)$. If we have access to a quantum S-AES oracle U_P that correctly maps the input state $|K\rangle$ to the ciphertext state $|C\rangle$, i.e., $U_P|K\rangle = |g_P(K)\rangle$, then the oracle O , as illustrated in Fig. 1, can be constructed accordingly.

The complexity of the S-AES oracle is decreased efficiently by some optimization improvements, and the overall quantum circuit is depicted in

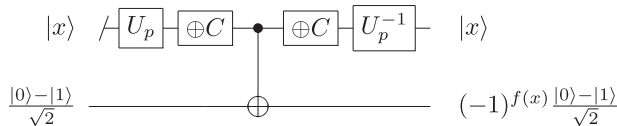


Fig. 1 | The schematic diagram of the oracle in the Grover-based attack. U_P is the quantum implementation of S-AES. $\oplus C$ signifies implementing an X gate for the i -th qubit if $C_i = 0$. The last auxiliary qubit undergoes a transformation from $(|0\rangle - |1\rangle)/\sqrt{2}$ to $(-1)^{f(x)}(|0\rangle - |1\rangle)/\sqrt{2}$ if the quantum system is in the state $|K_0\rangle$.

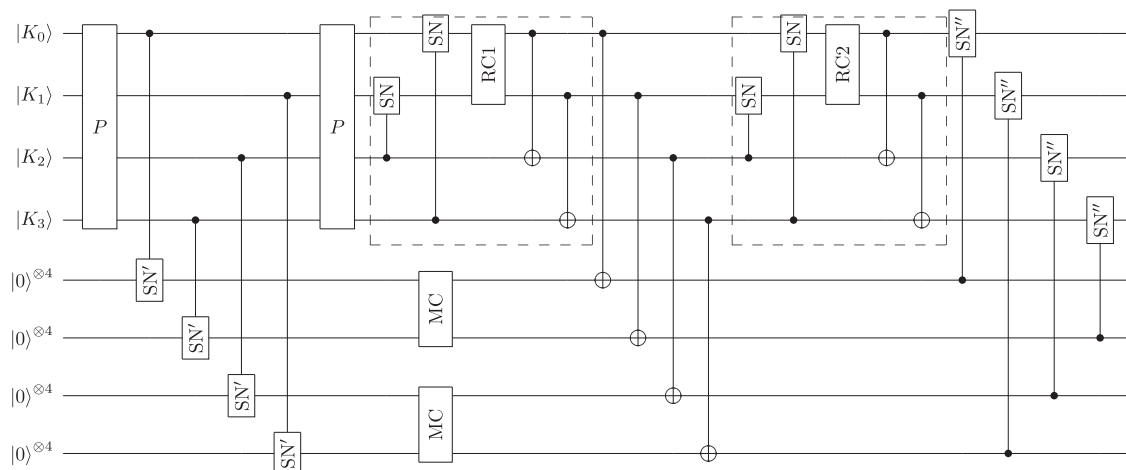


Fig. 2 | The quantum implementation of S-AES. For a 16-bit key $k_0 \sim k_{15}$, $K_0 = k_0k_1k_2k_3$, $K_1 = k_4k_5k_6k_7$, $K_2 = k_8k_9k_{10}k_{11}$, $K_3 = k_{12}k_{13}k_{14}k_{15}$. The ciphertext is encoded by the first 16 qubits as output. The dashed boxes execute the generation of SubKey-1 and SubKey-2, respectively. P encodes the known plaintext onto the corresponding

Fig. 2. The first 16-qubit is used to encode the key, while the last 16-qubit is initialized to $|0\rangle^{\otimes 16}$. The plaintext is encoded via an operation denoted by P . Specifically, when the m -th bit of the plaintext is 1, an X gate is applied to the corresponding m -th qubit; otherwise, no action is taken. Operations RC1 and RC2 are responsible for performing XOR operations on an 8-bit key, with RC1 utilizing the bit sequence 10000000 and RC2 utilizing 00110000, respectively. The ciphertext is encoded on the first 16-qubit through the S-AES oracle. It is worth noting that ShiftRows could be completed by redefining the order of qubits, so there is no extra operation.

The quantum circuit for SN is illustrated in Fig. 3, while the detailed explanations regarding the basis transformation and affine transformation are provided in Supplementary Note II. It is worth noting that for SN, the PN^{-1} operation is not required, and similarly, for SN', the NPA^{-1} operation is unnecessary. The quantum circuit for IO within SN', where the output qubits are in the state $|0\rangle^{\otimes 4}$, is shown in Fig. 4a, designed according to the sequence described in Eq. (1).

$$\begin{aligned} x_2 &= x_1 + x_2 & y_3 &= x_1 \times x_3 & x_4 &= y_4 + x_4 \\ y_4 &= x_2 \times x_4 & x_2 &= x_1 + x_2 & x_4 &= y_3 + x_4 \\ y_4 &= x_2 + y_4 & x_3 &= x_3 + x_4 & x_2 &= y_3 + x_2 \\ y_2 &= x_2 \times x_3 & x_2 &= y_3 + x_2 & y_2 &= x_2 \times x_3 \\ x_2 &= y_3 + x_2 & y_2 &= y_2 + x_4 & x_3 &= x_4 + x_3 \\ x_3 &= x_3 + y_2 & y_2 &= y_2 + y_3 + x_4 & y_1 &= x_4 \times y_2 \\ x_4 &= y_3 + x_4 & y_2 &= y_2 + x_4 & y_1 &= x_4 + y_1 \\ y_3 &= x_4 + y_3 & y_3 &= y_3 + y_1 \times y_4 & y_3 &= y_3 + x_1 + x_2 \\ y_1 &= y_1 + x_4 & y_1 &= y_1 + x_3 & x_3 &= x_3 + y_2 \\ x_4 &= x_4 + x_1 \times x_3 & y_3 &= y_3 + x_4 \end{aligned} \quad (1)$$

For the IO within SN and SN'', where the output qubits are in a random quantum state, we proposed an efficient quantum circuit that is depicted in Fig. 4b. As a result, 12 Toffoli gates and 14 CNOT gates are required in this operation.

The quantum circuit of MixColumn is provided in Supplementary Note II. As a critical component of the quantum S-AES Oracle, the implementation of both the MixColumn and SubNibble operations facilitates the straightforward construction of the quantum S-AES Oracle.

The quantum resources required to implement the S-AES oracle are summarized in Supplementary Table 1 of Supplementary Note I, while a comparative analysis is presented in Table 1. It is important to note that, in

qubits. RC1 and RC2 perform XOR operations with the known bit string. MC represents the MixColumns operation. SN, SN', and SN'' denote the SubNibble operations.

Table 1, a $\wedge_3 X$ gate is decomposed into four Toffoli gates. Since the Toffoli gate is a three-qubit non-Clifford gate, reducing the number of Toffoli gates will significantly decrease the overall quantum resource cost.

Grover-based attack

In the Grover-based attack, the initial state is prepared into a superposition state by Hadamard gates. The dimension of the space is 2^{16} , representing the 2^{16} possible keys. After $n = \frac{\pi}{4} \sqrt{N/M}$ Grover iterations, the target key will be found by measurement, where N is the number of items in the database, and M is the number of target keys.

The quantum circuit for the Grover iteration, along with its total resource cost, is detailed in Supplementary Note II. Assuming a single target ($M=1$), the required number of iterations is given by $n = \frac{\pi}{4} \sqrt{N/M} \approx 201$. As shown in Supplementary Table 2 of Supplementary Note II, the total gate count for the Grover-based attack amounts to 69,144 Toffoli gates, 157,584 CNOT gates, 7,236 Hadamard gates, and up to 36,582 Pauli-X gates.

What needs to be emphasized is that if Toffoli gates are decomposed into Clifford gates and T gates, the S-AES's Grover-based attack requires 572,448 CNOT gates, 484,008 T gates, 145,524 H gates, 69,144 S gates, and up to 36,582 X gates.

We simulate the S-AES's Grover-based attack using Qiskit, yielding impressive results. For example, we input the plaintext 0110 1111 0110 1011 and ciphertext 0000 0111 0011 1000 into the Grover-based attack program, performing 10^5 measurements. The output consists of the results 1010 0111

0011 1011 and 1010 0100 0101 1111, both of which are correct keys for the given plaintext and ciphertext. Further simulation results are presented in Table 2.

A VQAA without quantum implementation of symmetric cryptography

VQAA is based on the parameterized quantum circuit²². There are several key components in VQAA, including the classical optimization algorithms, ansatz, cost function, and the quantum implementation of symmetric cryptography (e.g., quantum AES oracle), as shown in Fig. 5a. Here, we propose an enhanced VQAA without quantum implementation of symmetric cryptography. The quantum AES oracle is replaced by the classical AES procedure, as shown in Fig. 5b.

We will demonstrate that the quantum AES oracle can be replaced by the classical AES procedure. Specifically, the procedure depicted in Fig. 5a is equivalent to that shown in Fig. 5b. The quantum AES oracle, composed of Toffoli gates, CNOT gates, and X gates, merely transforms one computational basis to another without generating any superposition between different computational bases. The input quantum state of the AES oracle can be represented as $|\psi(\theta)\rangle = \sum_i c_i |x_i\rangle$, assuming the AES procedure maps x_i to $y_i = f(x_i)$. After the AES oracle, the output state becomes $|\psi'(\theta)\rangle = \sum_i c_i |y_i\rangle$. If the relationship between x_i and y_i is a one-to-one mapping, the probability of obtaining y_i after implementing the quantum AES oracle is $|c_i|^2$. Similarly, the probability of x_i before implementing the quantum AES oracle is also $|c_i|^2$. Therefore, the y_i can be obtained with a probability of $|c_i|^2$ using the classical AES process. If there exist several x values, namely $x_{i(1)}, x_{i(2)}, \dots, x_{i(s)}$, all satisfying $f(x_{i(k)}) = y_i$, then the probability of obtaining y_i is $|c_{i(1)}|^2 + |c_{i(2)}|^2 + \dots + |c_{i(s)}|^2$, regardless of which procedure in Fig. 5 is chosen. The VQAA we proposed significantly reduces the depth of the quantum circuit.

The experiment results

In this section, we use S-AES as an example to demonstrate the validation of our VQAA proposal. Specifically, experiments for both the quantum S-AES

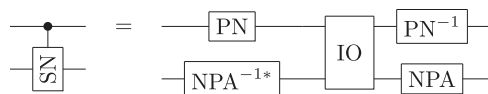
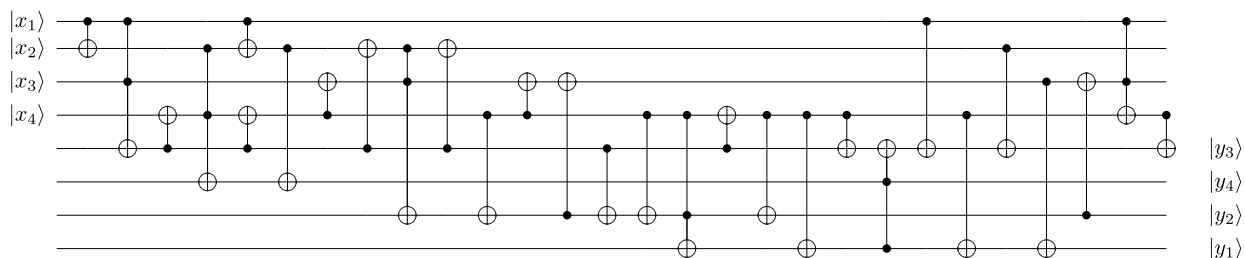
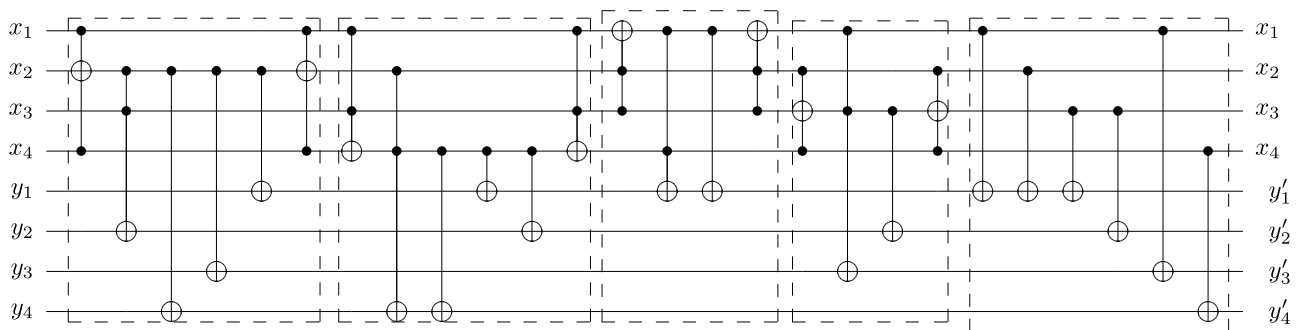


Fig. 3 | The SN circuit. IO represents the multiplicative inverse operation. PN represents the circuit for mapping polynomial basis to normal basis. NPA represents the combination circuit of mapping normal basis to polynomial basis and affine transformation. PN^{-1} is the inverse of PN. NPA^{-1} is the inverse of NPA.



(a)



(b)

Fig. 4 | Detailed quantum circuits of SN, SN', and SN''. a The IO circuit of SN'. b The IO circuit of SN and SN'.

oracle and the VQAA are conducted on the IBM superconducting platform²⁴.

Firstly, the performance of implementing the quantum S-AES oracle is demonstrated. Figure 6 shows the results of implementing three crucial operations, including XOR, MixColumn and multiplicative inverse operations. The first item in the bar graph of Fig. 6 represents the probability of the target state, while the remaining states are arranged in descending order of probability. Figure 6a and b display the results of the 16-qubit XOR and MixColumn operations. The target state was successfully obtained with a probability exceeding 0.5. For the SubNibble operation, which necessitates a

Table 1 | Quantum resource costs of different proposals for the S-AES oracle

Method	Qubits	Toffoli gate	CNOT	X gate
Ref. 8	72	192	568	10
Ref. 10	48	168	364	75
Ref. 15	32	160	144	35
This work	32	120	392	59

Table 2 | The simulation results of S-AES's Grover-based attack with the plaintext {0110 1111 0110 1011}

Ciphertext	Iteration	Measurement Result	Times
0010 1001 1000 1000	101	1011 0010 1100 1010	24856
		1000 0110 1010 0110	25146
		0011 1000 0101 1011	24892
		1100 0111 0110 1000	25084
		Others	22
0000 0111 0011 1000	142	1010 0111 0011 1011	50083
		0000 0111 0011 1000	49917
1110 0100 1001 1011	201	1111 1111 1000 0011	100000

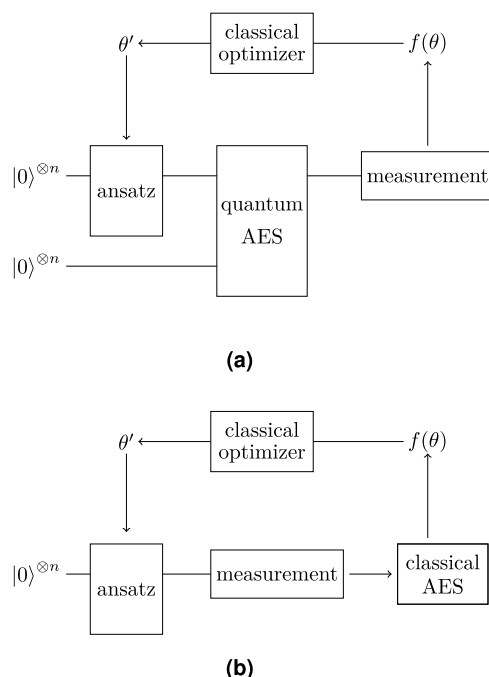


Fig. 5 | The procedure of VQAA for AES. a The procedure of VQAA in ref. 22. **b** Our enhanced procedure of VQAA is equivalent to a but with a shallower depth of the quantum circuit.

substantial number of Toffoli gates, Fig. 6c presents the experimental results of the multiplicative inverse operation in the SubNibble. The appearance probability of the target state is $P=0.11$.

In Fig. 7, the experimental performance of the multiplicative inverse operation is analyzed across various input configurations. The results demonstrate that the target state can be clearly distinguished based on the probability distribution.

The experimental results confirm that a quantum S-AES oracle can be implemented modularly on the IBM quantum platform. For a randomly selected pair of plaintext $P=1000\ 0110\ 0111\ 0110$ and key $K=1110\ 1100\ 1100\ 1111$, the highest probability output from each module was recorded and used as the input for the subsequent module. The agreement between simulation and hardware results confirms the reliability of the proposed implementation.

The VQAA, utilizing the single-layer configuration depicted in Fig. 10b, is tested on the IBM platform. The Nelder-Mead (N-M) method is employed for optimization, with the cutoff parameter initially set to $xerr=-80$, which is illustrated in Supplementary Note III. The plaintext is set to $P=1101\ 1111\ 0110\ 0100$, the ciphertext is set to $C=1111\ 0100\ 1011\ 0110$, and the corresponding key is $K=0101\ 0100\ 0100\ 1010$. The initial state is prepared to be close to the target state. By incorporating the parameter values generated during the experimental iterations into the simulation program, the variations in expectation values and fidelity throughout the iteration process are presented in Fig. 8. It demonstrates that the expectation value fluctuates downward and stabilizes with the iterations. The fidelity of the final quantum state exceeds 0.74, as calculated, and its probability distribution is shown in Fig. 9. The inset in Fig. 9 displays the probability distribution of the initial quantum state, with a fidelity of approximately 0.51. In both graphs, the first bar represents the probability of the target state, while the remaining elements are arranged in descending order of probability.

For the ciphertext $C=1111\ 0100\ 1011\ 0110$, the Hamiltonian in Eq. (10) exhibits eigenvalues $-120, -90, -64, -42, -24, -10, 0, 6, 8$, and 2 , with corresponding degeneracies of $32, 240, 1120, 3640, 8736, 16016, 22880$, and 12870 , respectively.

Discussion

In this paper, we present an efficient quantum S-AES oracle requiring 32 qubits, 120 Toffoli gates, and hundreds of Clifford gates, which can be utilized in Grover-based attacks. We implement simulations of S-AES's Grover-based attack using Qiskit and estimate the resource costs. Additionally, we propose a VQAA that reduces the depth of the quantum circuit. By reducing quantum resources using the above methods, we are able to demonstrate the quantum attack of S-AES on real superconducting quantum hardware. The results indicate the feasibility of cryptographic analysis on real quantum hardware.

For a parameterized quantum algorithm, our simulation results suggest that achieving the target state is challenging when the initial parameters are selected randomly. To address this, we propose a heuristic to explain why VQAA for S-AES appears to be inefficient. The function that maps plaintext and key to ciphertext is complex enough that, for one key K_i , the corresponding value of the cost function $E=E(C_i)=E(f(K_i))$ exhibits a random distribution. The average values $E(y_1)=\frac{1}{M}\sum_{i=1}^M E(K_i)$ and $E(y_2)=\frac{1}{M}\sum_{i=1}^M E(K'_i)$ are calculated after M times measurement according to parameters y_1 and y_2 . If $E(y_1)<E(y_2)$, the optimization will increase the superposition coefficients of the basis $|K_i\rangle$ and decrease the superposition coefficients of the basis $|K'_i\rangle$. However, for an ansatz that is unrelated to S-AES, the connection between $|K_i\rangle$ and the target basis $|K_0\rangle$ has no reason to be stronger than that between $|K'_i\rangle$ and $|K_0\rangle$. Therefore, for S-AES's VQAA, the key issue may be to construct an efficient ansatz based on the structure of S-AES.

Methods

In this section, we introduce the design for the IO circuit shown in Fig. 4b, which plays a critical role in reducing the quantum resource requirements

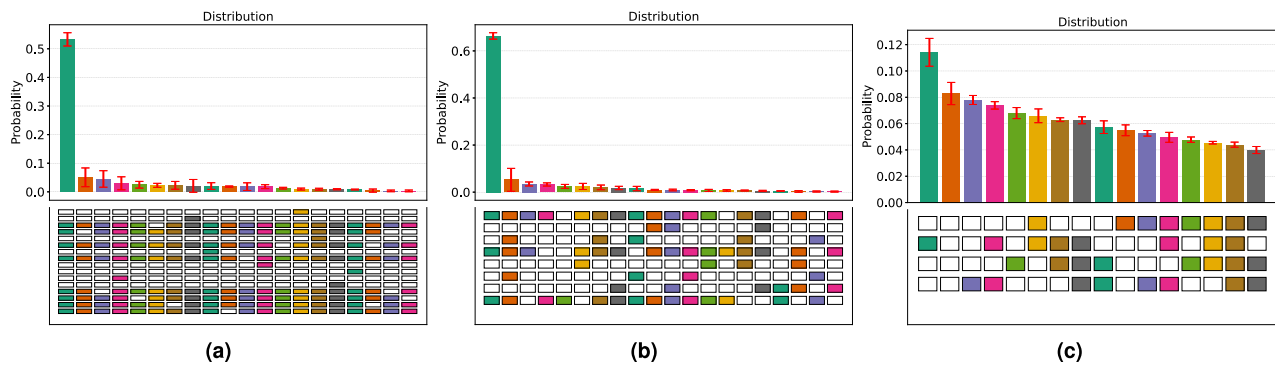


Fig. 6 | The experimental results of the quantum S-AES oracle on the IBM quantum platform. In each figure, the colored block represents logical “1”, while the white block denotes logical “0”. A column block below the x-axis represents an output of the experiment result. We repeated each quantum experiment three times, and in each repetition, we used 10,000 shots to sample the output distribution on the

IBM Q backend. **a** XOR: The circuit consists of 32 qubits, which are divided into two 16-qubit segments representing the key and plaintext, respectively, with their values selected at random. **b** MixColumn: The input state is $|10000111\rangle$. **c** SubNibble (multiplicative inverse part): The input state is $|1000\rangle$.

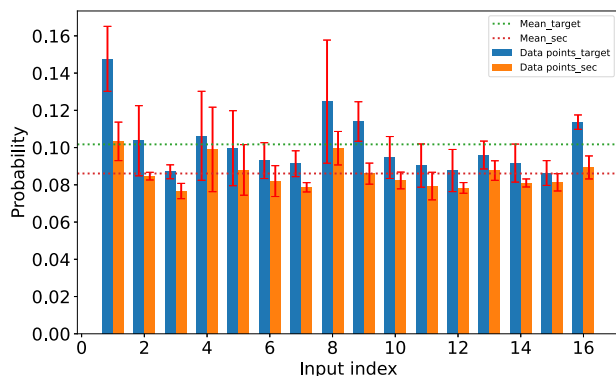


Fig. 7 | The probability distribution with error bars for the multiplicative inverse operation on the IBM quantum platform. The X-axis represents the input index, corresponding to the 16 different inputs ranging from “0000” to “1111”. The Y-axis represents the probability of the corresponding results. The blue bars indicate the probability of the target state, while the orange bars represent the probability of the most likely non-target state. The green dashed line denotes the average probability of the target state, and the red dashed line denotes the average probability of the most likely non-target state.

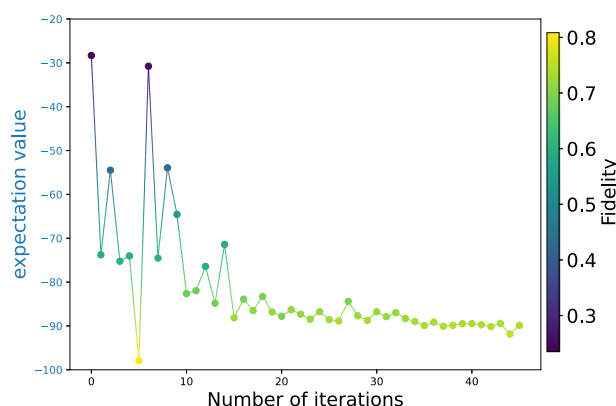


Fig. 8 | The experiment results of VQAA using the N-M method on the IBM platform. The Y-axis represents the expectation value of the cost function, and the color bar indicates the fidelity of the target state. The data points are obtained by taking the parameters generated at each iteration during the hardware execution, and then evaluating their corresponding cost function using an ideal quantum simulator.

for the quantum implementation of S-AES. Additionally, we describe the components and configuration of the VQAA.

An efficient design for SN and SN'

AES, the most widely used symmetric cryptography, supports plaintext length of $m = 128$ and key length of $m' = 128, 192$ and 256 , ensuring safe information transfer due to the sufficient key length. S-AES, a simplified version of AES, has $m = m' = 16$. Both the key and the plaintext are organized into a rectangular array of 2×2 nibbles. It serves as a toy model illustrating AES operations. The procedure of S-AES is provided in Supplementary Note I. Below is a brief introduction to the operations in S-AES.

AddRoundKey (ARK): It implements the XOR operation between the two inputs. For example, if the input are (a_1, a_2, a_3, a_4) and (b_1, b_2, b_3, b_4) , the output is $(a_1 \oplus b_1, a_2 \oplus b_2, a_3 \oplus b_3, a_4 \oplus b_4)$.

ShiftRows (SR): It holds the first row of the rectangular array and exchanges the second row.

$$\begin{pmatrix} N_0 & N_2 \\ N_1 & N_3 \end{pmatrix} \xrightarrow{\text{SR}} \begin{pmatrix} N_0 & N_2 \\ N_3 & N_1 \end{pmatrix} \quad (2)$$

MixColumns (MC): It performs a linear transformation for a column in the rectangular array. The matrix operation is as follows.

$$\begin{pmatrix} b_7 \\ b_6 \\ b_5 \\ b_4 \\ b_3 \\ b_2 \\ b_1 \\ b_0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_7 \\ a_6 \\ a_5 \\ a_4 \\ a_3 \\ a_2 \\ a_1 \\ a_0 \end{pmatrix} \quad (3)$$

SubNibbles (SN): It contains two operations, finding the multiplicative inverse in the finite field $\text{GF}(2^4)$ and an affine transformation. To reduce complexity, the multiplicative inverse operation is implemented on the normal basis. Specifically, SN, mapping $X_1X_2X_3X_4$ to $Y_1Y_2Y_3Y_4$, can be divided into three steps.

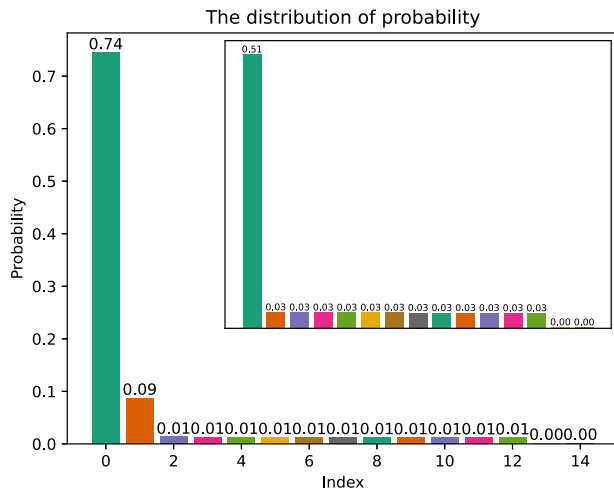


Fig. 9 | The probability distribution of the final quantum state. The main figure shows the probability distribution of the optimized quantum state, while the inset displays the probability distribution of the initial quantum state. In both figures, the first item represents the proportion of the target state.

Firstly, mapping $X_1X_2X_3X_4$ from the polynomial basis to the normal basis:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{pmatrix} \quad (4)$$

Secondly, finding the multiplicative inverse of $x_1x_2x_3x_4$ in $GF(2^4)$ on the normal basis:

$$\begin{aligned} y_1 &= x_2x_3x_4 + x_1x_3 + x_2x_3 + x_3 + x_4 \\ y_2 &= x_1x_3x_4 + x_1x_3 + x_2x_3 + x_2x_4 + x_4 \\ y_3 &= x_1x_2x_4 + x_1x_3 + x_1x_4 + x_1 + x_2 \\ y_4 &= x_1x_2x_3 + x_1x_3 + x_1x_4 + x_2x_4 + x_2 \end{aligned} \quad (5)$$

Finally, mapping $y_1y_2y_3y_4$ from normal basis to polynomial basis and performing the affine transformation, resulting in $Y_1Y_2Y_3Y_4$:

$$\begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} \oplus \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad (6)$$

It can be observed that Eq. (5) contains four 3rd-order terms ($x_2x_3x_4$, $x_1x_3x_4$, $x_1x_2x_4$, $x_1x_2x_3$) and four 2nd-order terms (x_1x_3 , x_2x_3 , x_2x_4 , x_1x_4). Therefore, one-to-one correspondences can be established between the 3rd-order terms and the 2nd-order terms. Eq. (7) presents one such corresponding relation:

$$\begin{aligned} x_1x_4 &\rightarrow x_1x_3x_4 \\ x_1x_3 &\rightarrow x_1x_2x_3 \\ x_2x_3 &\rightarrow x_2x_3x_4 \\ x_2x_4 &\rightarrow x_1x_2x_4 \end{aligned} \quad (7)$$

To begin with, the corresponding relation $x_1x_4 \rightarrow x_1x_3x_4$, the operation steps of IO within SN and SN² are presented in Eq. (8), and the corresponding quantum circuit is depicted in Fig. 4b. In Fig. 4b, the first Toffoli

gate implements x_1x_4 , the second Toffoli gate realizes $x_1x_3x_4$, and the third Toffoli gate is employed to restore the second qubit.

$$\begin{aligned} x_2 &= x_2 + x_1 \times x_4 & y_2 &= y_2 + x_2 \times x_3 & y_4 &= y_4 + x_2 \\ y_3 &= y_3 + x_2 & y_1 &= y_1 + x_2 & x_2 &= x_2 + x_1 \times x_4 \\ x_4 &= x_4 + x_1 \times x_3 & y_4 &= y_4 + x_2 \times x_4 & y_4 &= y_4 + x_4 \\ y_1 &= y_1 + x_4 & y_2 &= y_2 + x_4 & x_4 &= x_4 + x_1 \times x_3 \\ x_1 &= x_1 + x_2 \times x_3 & y_1 &= y_1 + x_1 \times x_4 & y_1 &= y_1 + x_1 \\ x_1 &= x_1 + x_2 \times x_3 & x_3 &= x_3 + x_2 \times x_4 & y_3 &= y_3 + x_1 \times x_3 \\ y_2 &= y_2 + x_3 & x_3 &= x_3 + x_2 \times x_4 & y_1 &= y_1 + x_1 \\ y_1 &= y_1 + x_2 & y_1 &= y_1 + x_3 & y_2 &= y_2 + x_3 \\ y_3 &= y_3 + x_1 & y_4 &= y_4 + x_4 \end{aligned} \quad (8)$$

The first dashed box in Fig. 4b completes the corresponding relation $x_1x_4 \rightarrow x_1x_3x_4$. As a result, 12 Toffoli gates and 14 CNOT gates are required for the entire operation.

Considering the corresponding relations and the operation order, eight cases need to be analyzed. The results indicate that the quantum resource cost remains identical across all cases. More importantly, this design concept can be applied to the quantum implementation of AES, thereby reducing the overall consumption of quantum resources. The details are discussed in Supplementary Note IV.

The components of VQAA

For classical optimization, commonly used algorithms include gradient descent and the Nelder-Mead (NM) method²⁵, among others. The specific optimization parameters may require adjustment depending on the problem. In our implementation of VQAA, we employ the Nelder-Mead method to reduce the likelihood of converging to a local minimum. The optimization process is restarted if the conditions $f(x_i) - f(x_0) < \delta$ and $f(x_0) > \epsilon$ are satisfied, where f is the cost function (corresponding to the expectation value of the Hamiltonian), x_0 is the current parameter, x_i is the last parameter and ϵ and δ are predefined thresholds. Furthermore, if $f(x_0) < \epsilon$, the iteration is terminated. It is important to note that the process may terminate before reaching $f(x_0) < \epsilon$ if the measurement result successfully matches the known ciphertext.

For the designing of ansatz, we provide two ansatzes used in VQAA as shown in Fig. 10, where RSG gates in Fig. 10a implement the following unitary:

$$\text{RBS}(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta) & \sin(\theta) & 0 \\ 0 & -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (9)$$

We choose them as the ansatzes for the SAES circuit based on the following considerations: (1) The gate complexity of the ansatzes in Fig. 10a and b are $O(n \log n)$ and $O(n)$, respectively, making them possible to be executed on real quantum devices. (2) Both ansatzes can prepare the initial state $|0\rangle^{\otimes 16}$ into any computational basis state. Therefore, the parameterized circuit can, under different parameter configurations, prepare a final quantum state corresponding to any possible key in the computational basis. This ensures the solution space is complete - no valid key is excluded from being represented by the circuit's output state. (3) The circuit in Fig. 10a implements an all-to-all entanglement pattern. The 2-layer $R_y(\theta)$ -CNOT circuit in Fig. 10b also creates correlations between all qubits.

For the cost function, it is defined as the expectation of Hamiltonian, namely, $E = \langle \psi | H | \psi \rangle$, where $|\psi\rangle$ is the quantum state. A suitable

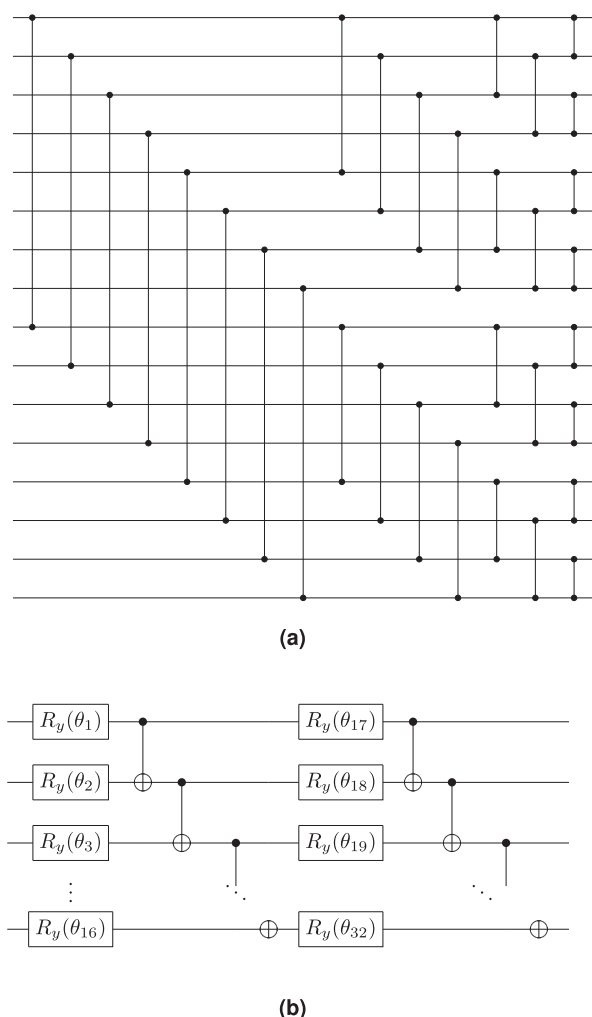


Fig. 10 | Two ansatz used in the VQAA for S-AES. a The ansatz consisting of the RBS gates⁴⁶, which leaves the basis states $|00\rangle$ and $|11\rangle$ unchanged, maps $|01\rangle$ to $\cos \theta|01\rangle - \sin \theta|10\rangle$, and $|10\rangle$ to $\sin \theta|01\rangle + \cos \theta|10\rangle$. **b** A simple ansatz consisting of $R_y(\theta)$ and CNOT gates.

Hamiltonian based on the known ciphertext C is constructed.

$$H = \sum_i t_i Z_i + \sum_{ij} \omega_{ij} Z_i Z_j \quad (10)$$

where $t_i = 1$ if the i -th bit of ciphertext $C_i = 1$ and $t_i = -1$ if $C_i = 0$, $\omega_{ij} = 1$ if $C_i \neq C_j$ and $\omega_{ij} = -1$ if $C_i = C_j$.

Data availability

All relevant data and figures supporting the main conclusions of the document are available on request. Please refer to Zeguo Wang at wangzeguo@quantumsc.cn.

Code availability

All relevant code supporting the document is available upon request. Please refer to Zeguo Wang at wangzeguo@quantumsc.cn.

Received: 14 November 2024; Accepted: 3 September 2025;

Published online: 03 October 2025

References

1. Rivest, R. L., Shamir, A. & Adleman, L. A method for obtaining digital signatures and public-key cryptosystems. *Commun. ACM* **21**, 120–126 (1978).

2. Shor, P. W. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM Rev.* **41**, 303–332 (1999).
3. Monz, T. et al. Realization of a scalable Shor algorithm. *Science* **351**, 1068–1070 (2016).
4. Ahnefeld, F., Theurer, T., Egloff, D., Matera, J. M. & Plenio, M. B. Coherence as a resource for Shor's algorithm. *Phys. Rev. Lett.* **129**, 120501 (2022).
5. Grover, L. K. A fast quantum mechanical algorithm for database search. In *Proceedings of the 28th Annual ACM Symposium on Theory of Computing*, 212–219 (1996).
6. Long, G.-L. Grover algorithm with zero theoretical failure rate. *Phys. Rev. A* **64**, 022307 (2001).
7. Anikeeva, G. et al. Number partitioning with Grover's algorithm in central spin systems. *PRX Quantum* **2**, 020319 (2021).
8. Almazrooie, M., Samsudin, A., Abdullah, R. & Mutter, K. N. Quantum reversible circuit of AES-128. *Quantum Inf. Process.* **17**, 1–30 (2018).
9. Langenberg, B., Pham, H. & Steinwandt, R. Reducing the cost of implementing the advanced encryption standard as a quantum circuit. *IEEE Trans. Quantum Eng.* **1**, 1–12 (2020).
10. Wang, Z.-G., Wei, S.-J. & Long, G.-L. A quantum circuit design of AES requiring fewer quantum qubits and gate operations. *Front. Phys.* **17**, 41501 (2022).
11. Li, Z. et al. Novel quantum circuit implementation of the Advanced Encryption Standard with low costs. *Sci. China Phys., Mech. Astron.* **65**, 290311 (2022).
12. Mandal, S., Anand, R., Rahman, M., Sarkar, S. & Isobe, T. Implementing Grover's on AES-based aead schemes. *Sci. Rep.* **14**, 21105 (2024).
13. Musa, M. A., Schaefer, E. F. & Wedig, S. A simplified AES algorithm and its linear and differential cryptanalyses. *Cryptologia* **27**, 148–177 (2003).
14. Almazrooie, M., Abdullah, R., Samsudin, A. & Mutter, K. N. Quantum Grover attack on the simplified AES. In *Proceedings of the 2018 7th International Conference on Software and Computer Applications*, 204–211 (2018).
15. Jang, K.-B., Song, G.-J., Kim, H.-J. & Seo, H.-J. Grover on simplified aes. In *2021 IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia)*, 1–4 (IEEE, 2021).
16. Cerezo, M. et al. Variational quantum algorithms. *Nat. Rev. Phys.* **3**, 625–644 (2021).
17. Peruzzo, A. et al. A variational eigenvalue solver on a photonic quantum processor. *Nat. Commun.* **5**, 4213 (2014).
18. Farhi, E., Goldstone, J. & Gutmann, S. A quantum approximate optimization algorithm. arXiv preprint arXiv:1411.4028 (2014).
19. Wei, S., Chen, Y., Zhou, Z. & Long, G. A quantum convolutional neural network on NISQ devices. *AAPPS Bull.* **32**, 1–11 (2022).
20. Zhou, Z.-r., Li, H. & Long, G.-L. Variational quantum algorithm for node embedding. *Fundam. Res.* **4**, 845–850 (2024).
21. Heyraud, V., Li, Z., Donatella, K., Le Boité, A. & Ciuti, C. Efficient estimation of trainability for variational quantum circuits. *PRX Quantum* **4**, 040335 (2023).
22. Wang, Z., Wei, S., Long, G.-L. & Hanzo, L. Variational quantum attacks threaten advanced encryption standard-based symmetric cryptography. *Sci. China Inf. Sci.* **65**, 200503 (2022).
23. Cross, A. The IBM Q experience and Qiskit open-source quantum computing software. In *APS March meeting abstracts*, vol. 2018, L58–003 (2018).
24. Platform, I. Q. <https://quantum-computing.ibm.com>.
25. Nelder, J. A. & Mead, R. A simplex method for function minimization. *Comput. J.* **7**, 308–313 (1965).
26. Cherrat, E. A. et al. Quantum vision transformers. *Quantum* **8**, 1265 (2024).

Acknowledgements

Z.W. acknowledges support from the Research Initiation Fund of the Quantum Science Center of the Guangdong-Hong Kong-Macao Greater Bay Area (Grant No. QD2305001). K.W. was supported in part by the same

fund (Grant No. QD2305001). S.W. acknowledges the Beijing Nova Program (Grants No. 20230484345 and 20240484609). We also acknowledge support from the National Natural Science Foundation of China (Grant No. 62471046).

Author contributions

Z.G.W. and M.X.Z. developed the theoretical model, performed simulations, and analyzed the experimental data. Z.G.W., M.X.Z., and S.J.W. designed the experimental protocol. J.W.W. conducted the experiments on the IBM platform. S.J.W. and G.L.L. provided supervision and guidance throughout the project. All authors contributed to the discussion of results and participated in the writing and revision of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at

<https://doi.org/10.1038/s41534-025-01106-w>.

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